

Consumer Valuation of Powertrain Type, Range, and Operating Cost in Used Vehicle Markets: A Discrete Choice Experiment

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Background

Used vehicles account for roughly 70% of U.S. vehicle transactions, yet most alternative fuel vehicle (AFV) research focuses on new-vehicle buyers. This study estimates willingness to pay (WTP) for powertrain type and other attributes among prospective U.S. used-vehicle buyers and compares WTP with real-market used-vehicle prices.

Data and Method

An online survey (Dynata and Prolific, December 2025–April 2026) of U.S. adults planning to buy a used car or SUV within two years yielded 3,072 respondents. Each completed six discrete choice tasks varying powertrain (conventional, gas hybrid [HEV], battery electric [BEV]), price, operating cost, BEV range, model year, and mileage. Using a qualified subsample of 2,254 respondents, we estimated six mixed logit models, stratified by car/SUV segment and then by budget tier (low/high), in WTP space. Simulated net WTP was also compared with real-market price premiums for matched used vehicle pairs (AFV vs conventional).

Results

Mean BEV utility is negative and highly significant in all models (-\$10.4k to -\$26.1k), with large standard deviations indicating within-segment heterogeneity. HEVs carry a significant premium only among high-budget buyers (about \$4,200 for cars, \$2,300 for SUVs). The BEV penalty narrows sharply with range, from about -\$8,200 to -\$18,800 at 100 miles to -\$2,600 to -\$11,700 at 300 miles (Figure 1); only high-budget car buyers considering a 300-mile BEV are statistically indistinguishable from the conventional baseline. Mileage, age, and operating cost matter less than powertrain and range. Observed used-BEV premiums exceed the WTP-implied parity requirement in every matched pair (by about \$6,100 for the high-budget car, \$15,700 for the low-budget car, and \$16,900 for the low-budget SUV), while HEV results are mixed. The former \$4,000 used clean vehicle credit could cover the gap for high-budget car buyers considering longer-range BEVs, but its \$25,000 price cap excludes most longer-range used BEVs. These results favor segment-aware policy, including non-price instruments such as battery health disclosure.

Willingness to Pay for Vehicle Attributes

By vehicle segment and budget tier; error bars show 95% confidence intervals

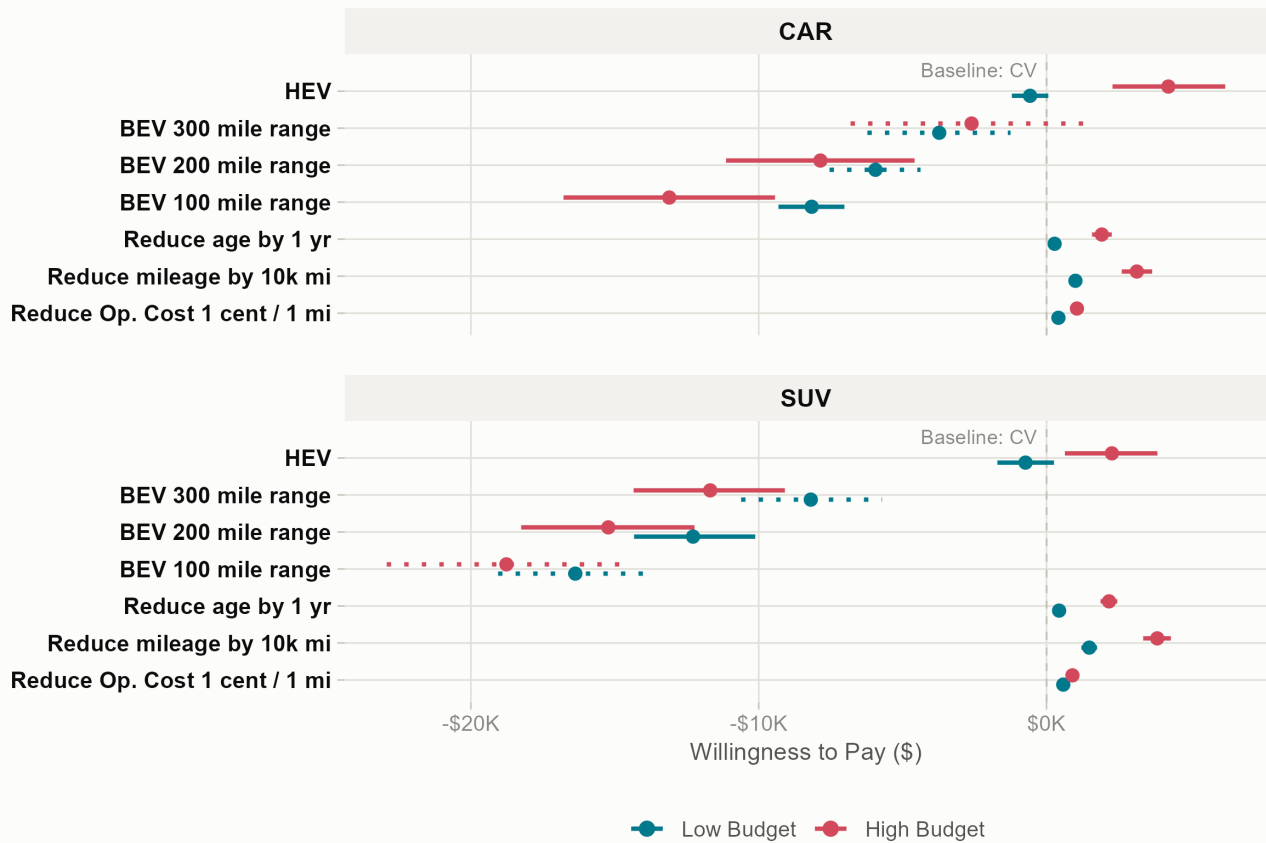


Figure 1: Willingness to pay for vehicle attributes, by vehicle segment and budget tier (95% confidence intervals shown; values in thousands of dollars). For the BEV range configurations, solid intervals mark WTP calculated within the range shown to that segment and dotted intervals mark WTP extrapolated beyond it.

Consumer Valuation of Battery Health Attributes in Used Battery Electric Vehicle Markets: A Discrete Choice Experiment

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Background

The growth of the used BEV market extends access to middle- and lower-income households, supports a resilient circular economy by extending vehicle and battery lifetimes, and underpins the residual value that makes new BEV leases financially viable. However, buyers of used battery electric vehicles (BEVs) face substantial uncertainty about battery condition at the point of sale. The information gap suppresses demand and prevents efficient pricing in the secondary market. This paper delivers the first willingness-to-pay (WTP) estimates for battery health attributes in the U.S. used BEV market.

Data and Method

We administered an online discrete choice experiment (December 2025–April 2026) to 3,072 U.S. adults intending to purchase a used car or SUV within two years. Each respondent completed six choice tasks among three hypothetical used BEVs or a no-choice option, varying mileage, price, electric range at Year 3, annual battery degradation rate, and battery refurbishment history. The design was stratified by body type and budget, and half the sample received extended battery cost and warranty information. We estimated a mixed logit model and a six-class Latent Class Choice Model on 2,916 respondents to reveal the WTP heterogeneity among consumers.

Results

At the aggregate level, consumers value electric range, penalize mileage and battery degradation, and discount refurbishment history, but the preference heterogeneity is substantial. The LCCM results suggest that consumers fall into six segments that differ markedly in their sensitivity to battery quality signals, their price responsiveness, and their market engagement. Classes 1 and 2 (43.5% combined) prioritize battery health with a high premium for vehicle range and clear refurbishment penalties, and are the primary market for certified battery-health disclosure. Class 3 (16.6%) is BEV-engaged but responds to range, not refurbishment history. Class 4 (12.8%) is price-constrained and largely indifferent to battery quality signals, while Class 5 (8.7%) represents consumers not yet viable in the used BEV market. The last class (18.5%) consists of low-engagement, attribute non-attendance respondents. The attached figure (Figure 2) shows more details on each class. These findings show the potential for battery-health disclosure and certification to serve as a market-transparency signal in used BEV transactions. For dealers and OEMs, state of health, degradation history, and remaining warranty coverage could command measurable price premiums from the quality-sensitive majority.



Figure 2: WTP and class profile summary.